INCH-POUND

MIL-PRF-1/781E
1 May 2007
SUPERSEDING
MIL-E-1/781D
5 May 1975

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 6D4

INACTIVE FOR NEW DESIGN
AFTER 30 APRIL 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for acquiring the electron tube described herein shall consist of this specification and MIL-PRF-1.

DESCRIPTION: Triode, noise generator.

Mounting position: Any.

Weight: 0.3 ounce (8.5 grams) nominal.

Base: E7-1 (EIA).

Envelope: T5-1/2 (5-2).

Pin connections:

Pin no.:	- - -	1	2	3	4	5	6	7
Element:	- - -	g	nc	h	h	k	nc	a

ABSOLUTE RATINGS:

Parameter:	Ef	Ebb	epx	epv	Ib	ib	tk	Ec1
Unit:	V	V dc	v	v	mA dc	a	sec	V dc
Maximum:	6.93	250	350	350	25	0.110	---	-150
Minimum:	5.67	---	---	---	---	---	30	---
Test conditions:	6.3	125	---	---	---	---	---	---

ABSOLUTE RATINGS:

Parameter:	Rhk	Rg	Rp	Rk	Ehk	Du	TA
Unit:	MegΩ	MegΩ	Ohms	Ohms	V	%	°C
Maximum:	---	---	---	---	-110	0.75	+90
Minimum:	---	---	---	---	---	---	-55
Test conditions:	1.0	0.5	650	4,000	---	---	---

GENERAL:

Qualification - Not required.

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TABLE I. Requirements or tests.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance level (see note 6)	Symbol	Limits		Unit
						Min	Max	
	Conformance inspection, part 1							
- - -	Grid-cathode voltage	1	Ec = -20 V dc; Rhk = 0	0.65	Egk	- - -	2.0	V dc
3241	Heater current	- - -		0.65	If	230	270	mA
3201	Critical grid voltage for conduction (1)	4		0.65	Ec	-11.0	-14.0	V dc
3204	Voltage drop	- - -	Rb/Ib = 100 mA dc	0.65	Etd	- - -	18	V dc
	Conformance inspection, part 2							
1336	Heater-cathode leakage	- - -	Ehk = -100 V dc	- - -	Ihk	---	15	μA dc
3201	Critical grid voltage for conduction (2)	- - -	Ebb = 50 V dc	- - -	Ec	-5.0	-7.0	V dc
3201	Critical grid voltage for conduction (3)	- - -	Ebb = 300 V dc	- - -	Ec	-21	-31	V dc
- - -	Noise output (1)	2	Ebb = 300 V dc; Rg = 0; Rp = 56,000 ohms	6.5	- - -	- - -	- - -	- - -
- --	Noise output (2)	3	Ebb = 250 V dc; Rg = Rk = 0; Rp = 0.33 MegΩ	- - -	Output	10	- - -	v
	Conformance inspection, part 3							
- - -	Life test	- - -	Group A; Ebb = 250 V dc; Ec = -20 V dc; Rp = 5,000 ohms; Rhk = disconnected; Ehk = 110 V; t = 500 hours	- - -	- - -	- - -	- - -	- - -
- - -	Life-test end points:							
3201	Critical grid voltage for conduction (1)	- - -		- - -	Ec	-9.5	-15.5	V dc
- - -	Noise output (2)	3		- - -	Output	9.0	- - -	v
1031	Low-frequency vibration	5	No voltages applied	- - -	- - -	- - -	- - -	- - -

See footnotes at top of next page.

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TABLE I. Requirements or tests - Continued.

NOTES:

1. Voltage measured across specified grid resistor.

2. The tube shall be placed in the circuit shown (see figure 1) in a constant magnetic field of 375 ± 10 percent gausses perpendicular to the normal electron path. The direction of the magnetic field shall be such as to deflect the electron beam toward the top of the tube (north pole of magnet at pin No. 7). The noise voltage measured at the output of the 1,000 Hz bandwidth filter shall not be less than the limits specified below for the various specified frequencies:

Frequency, MHz	Minimum noise voltage, μV rms
0.1	10,000
0.2	14,000
0.5	25,000
1.0	22,000
2.0	7,000
5.0	500
10.0	70

3. The tube shall be placed in the circuit shown (see figure 2) in a constant magnetic field of 375 ± 20 percent gausses which is perpendicular to the normal electron path. The direction of the magnetic field shall be such as to deflect the electron beam toward the top of the tube. The noise voltage measured at the anode of the tube and across the output of the circuit shall not be less than the specified limit in peak-to-peak volts. The oscilloscope used for noise amplitude measurement shall have a 3 dB video bandwidth extending to at least 4 MHz.

4. This test shall be the first test performed at the conclusion of the holding period.

5. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. An accept on zero defect (see note 6) sampling plan shall be used, with the sample of three tubes with an acceptance number of zero. In the event of a failure, the test will be made as a part of conformance inspection, part 2 with an acceptance level (see note 6) of 6.5. The regular "12-calendar month" sampling plan shall be reinstated after three consecutive samples have been accepted.

6. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

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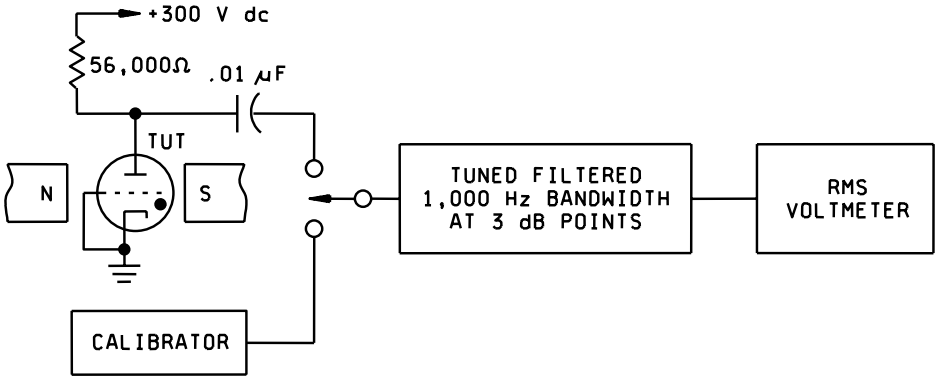

FIGURE 1. Noise voltage test circuit.

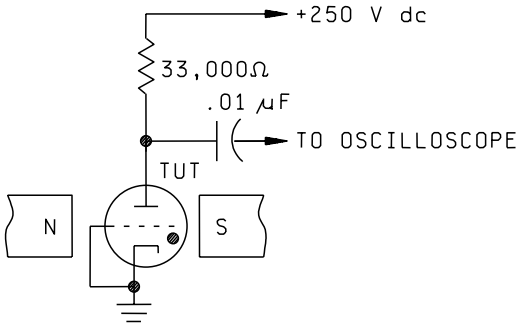

FIGURE 2. Noise voltage test circuit.

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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 99

Preparing activity:
DLA - CC

(Project 5960-2007-019)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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